Xiaoyu Zheng

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EDUCATION

Harbin Institute of Technology *08/2021-06/2025*

- B.Eng. Microelectronics Science & Engineering

University of Southampton *09/2025-09/2026*

- MSc. Microelectronics Systems Design

Nanjing University *09/2026-06/2031*

- Researcher && Ph.D. Digital Integrated Circuits & Hardware-Software Co-design

PUBLICATIONS

- X. Zheng, Z. Guo, C. Zhou#, "A Unified Function Processor with Integer Arithmetic Based on Piecewise Chebyshev Polynomial Approximation", *IEEE ISCAS*, 2026 && *TCAS-I* in preparation.
- Z. Qiao, X. Zheng, S. Taravati#, "Physical Neural Computing Enabled by a Dynamic Array of Interconnected Multi-Fed Patch Resonators", *EurAAP European Conference on Antennas and Propagation,* 2027, Manuscript under review.
- X. Zheng, K. Zhou, C. Zhou#, "Voxel-Based Point Cloud Processing Architecture for Hardware Acceleration", *IEEE Transactions on Circuits and Systems II (TCAS-II) && DAC,* 2027, in preparation.

RESEARCH EXPERIENCES

Hybrid Physical Neural Computing with Programmable Metasurfaces *03/2026-10/2026*

- Conceived and developed a physical neural computing architecture, including algorithm design, hardware and software implementation, and behavioral modelling;
- Designed and fabricated the PCB prototype, implemented the FPGA-based hardware control system, and built a complete closed-loop framework that supports on-chip inference with PC model training for system integration.

A Unified Function Processor with Integer Arithmetic Based on Piecewise Chebyshev Polynomial Approximation *07/2025-10/2025*

- Proposed the Quantization-Aware Polynomial Approximation Algorithm (QADP) based on Chebyshev, integrating segmentation and quantization via a two-layer dynamic programming framework;
- Built a configurable Python toolchain for design-space exploration and error–resource trade-off analysis.

Voxel-based Point Cloud Processing for Hardware Acceleration *09/2024-11/2026*

- Designed a point cloud sampling and neighbor-search algorithm based on voxel partitioning**,** replacing traditional FPS/KNN distance calculations with voxel index lookups;
- Reduced explicit distance computations by leveraging voxel index mapping.

PROJECT EXPERIENCES

Volans LLM Chip (Under the supervision of Dr. Changchun Zhou)

Team Member (NPU Subsystem Microarchitecture Co- designer and Developer) *02/2026-Present*

- Co-designed an 8-stage AXI-Stream-based microarchitecture with stage-level functional partitioning;
- Designing and implementing multiple modules including computing and AXI4 input/output interface at RTL level.

Fornax Diffusion Chip (Under the supervision of Dr. Changchun Zhou)

Team Member (OCI Module Developer, UFP Module Co-developer) *12/2025-01/2026*

- Designed the OCI (output reversely compresses input) and UFP(unified non-linear functions calculation) module, including module-level instruction registers configuration, micro-architecture design, dataflow, and RTL implementation;
- Completed module-level simulation and functional verification, finally integrated the module with CCU/AXI subsystems within the SoC.

Research on Intelligent Control and Application of Landscape Lighting in Complex Environments

Team Member of Controller Embedded Systems Part (Provincial-level Project) *10/2024-06/2025*

- Developed a dynamic audio-responsive lighting algorithm;
- Implemented Modbus protocol, ADC acquisition, and RS485 communication on STM32 platform, used in Harbin Ice and Snow World 2025.

SKILLS

- **Programming:** Verilog, SystemVerilog, C, Python
- **Digital ASIC Implementation**: Basic Full RTL-to-GDSII flow
- **FPGA & Embedded & Simulation:** Quartus, ModelSim, MATLAB, Altium Designer